



Public Land Mobile Network (PLMN)



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A Public Land Mobile Network refers to the entire infrastructure and system that allows mobile devices (such as cell phones and tablets) to communicate wirelessly within a designated coverage area.

Components of PLMN

- **Base Stations (Cell Towers):**
Transmit and receive signals to and from mobile devices within their coverage area.
- **Mobile Switching Center (MSC):**
Manages the connection between mobile devices and facilitates call routing.
- **Home Location Register (HLR):**
Stores subscriber information and handles tasks related to authentication and call routing.

- **Visitor Location Register (VLR):**
Maintains temporary subscriber information for devices currently within the coverage area.

Cellular Structure:

- PLMN is often organized into cells, each served by a base station. The entire coverage area is divided into these cells, allowing for efficient use of frequency spectrum and accommodating a large number of users.

Coverage Areas:

- PLMNs cover a specific geographical area, and multiple PLMNs collectively provide extensive coverage, allowing users to communicate seamlessly while moving within the network.

Network Operators:

- PLMNs are operated by network operators, which can be telecommunication companies or service providers. These operators are responsible for the deployment, maintenance, and management of the network infrastructure.

International Roaming:

- PLMNs support international roaming, allowing users to use their mobile devices in different countries. This is made possible through agreements and partnerships between network operators.

Communication Services:

- PLMNs provide a range of communication services, including voice calls, text messaging (SMS), multimedia messaging (MMS), and mobile data services.

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